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Trading Performance & Market Sentiment Analysis Report

1. Executive Summary

This analysis evaluates the relationship between the **Bitcoin Fear/Greed Index** and trader performance using historical data from the **Hyperliquid** platform. The findings suggest that market sentiment is a significant predictor of both profitability (Profit Factor) and risk appetite (Notional Value). Strategic patterns were identified in "Smart Money" behaviour, specifically that the most profitable traders capitalise on periods of "Extreme Greed" with balanced, non-directional strategies, while the general population tends toward more aggressive directional bias.

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2. Methodology

- **Data Merging:** Historical trading logs were standardised and merged with daily Bitcoin Sentiment classifications using a common date key. Missing sentiment values were forward-filled to ensure data continuity.
- **Performance Metrics:** The analysis focused on **Win Rate (%)** and **Profit Factor** (Gross Profits / Gross Losses) as primary indicators of performance.
- **Risk Metrics: Notional Value** (Size × Price) was used to measure capital commitment and risk appetite across sentiment levels.
- **Segmentation:** Accounts were split into the **Top 10%** (most profitable) and the **Bottom 90%** to compare "Smart Money" tactics against general market participants.

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3. Key Findings & Analysis

3.1 Sentiment vs. Profitability

Periods of **Greed** yielded the highest Profit Factor (7.23), while **Extreme Greed** resulted in the highest Win Rate (49.01%). In contrast, **Neutral** market sentiment proved the most difficult environment for traders, showing both the lowest win rates and profit factors.

Sentiment-Based Profitability Analysis:

classification	Win Rate (%)	Profit Factor
Extreme Greed	49.008905	3.218522
Fear	40.408625	4.088960
Greed	44.647138	7.227706
Neutral	31.718247	1.964129

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Sentiment Classification	Win Rate (%)	Profit Factor
Extreme Greed	49.01	3.22
Greed	44.65	7.23
Fear	40.41	4.09
Neutral	31.72	1.96

3.2 Risk Appetite (Capital Commitment)

Traders deploy significantly more capital during sentiment extremes. The average **Notional Value** peaked during periods of **Fear** (6,307) and **Extreme Greed** (5,660), nearly doubling the capital committed during Greed or Neutral phases. This indicates that traders increase their conviction (or "gambling" behaviour) when the market is at its most emotional.

3.3 Asset Sensitivity (BTC vs. Altcoins)

- **Altcoins:** Showed a consistently higher win rate during **Extreme Greed** (54.69%) compared to BTC.
- **Bitcoin (BTC):** While having lower win rates, BTC showed astronomical **Profit Factors** during **Greed (8.32)** and **Neutral (20.28)** periods. This suggests that while BTC winning trades are less frequent, they are substantially larger, offering high-reward opportunities for patient traders.

Asset Sensitivity Analysis (Profitability by Sentiment for BTC vs. Altcoins):

classification	BTC Win Rate (%)	BTC Profit Factor	Altcoin Win Rate (%)	Altcoin Profit Factor
Extreme Greed	27.429359	0.471754	54.690619	3.847735
Fear	35.733357	2.857418	41.153278	4.371961
Greed	33.710309	8.321293	45.386878	7.196335
Neutral	34.562212	20.276797	31.629116	1.903431

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4. "Smart Money" vs. Retail Behavior

A comparative analysis between the **Top 10%** of profitable accounts and the **Bottom 90%** revealed distinct strategic differences:

- **Opportunism:** The Top 10% are significantly more active during **Extreme Greed**, where 7.74% of their total trades occur, compared to only 2.27% for the Bottom 90%.
- **Directional Neutrality:** During Extreme Greed, top-performing traders maintain a highly balanced **Long/Short Ratio (0.97)**. The Bottom 90% tend to have more unbalanced ratios (0.92), suggesting that top traders utilize these periods for hedging or taking both sides of volatility rather than just "chasing" the trend.

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5. Visual Insights Summary

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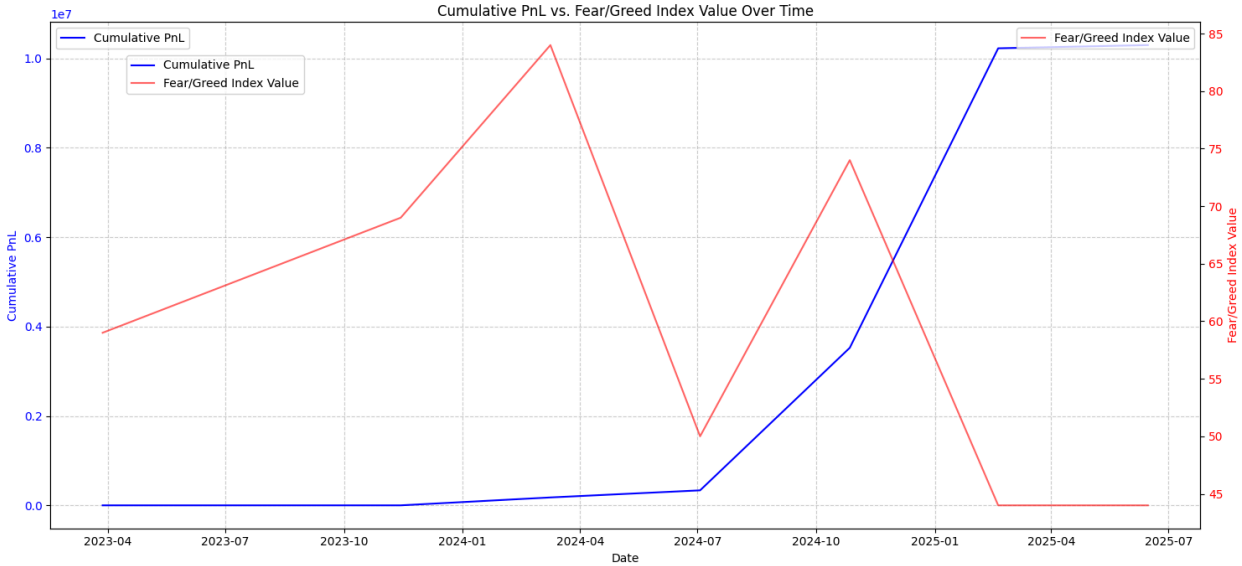
The following visualisations were generated to support these findings:

- **Cumulative PnL vs. Sentiment:** A dual-axis plot illustrating that equity growth often accelerates following shifts out of "Fear."



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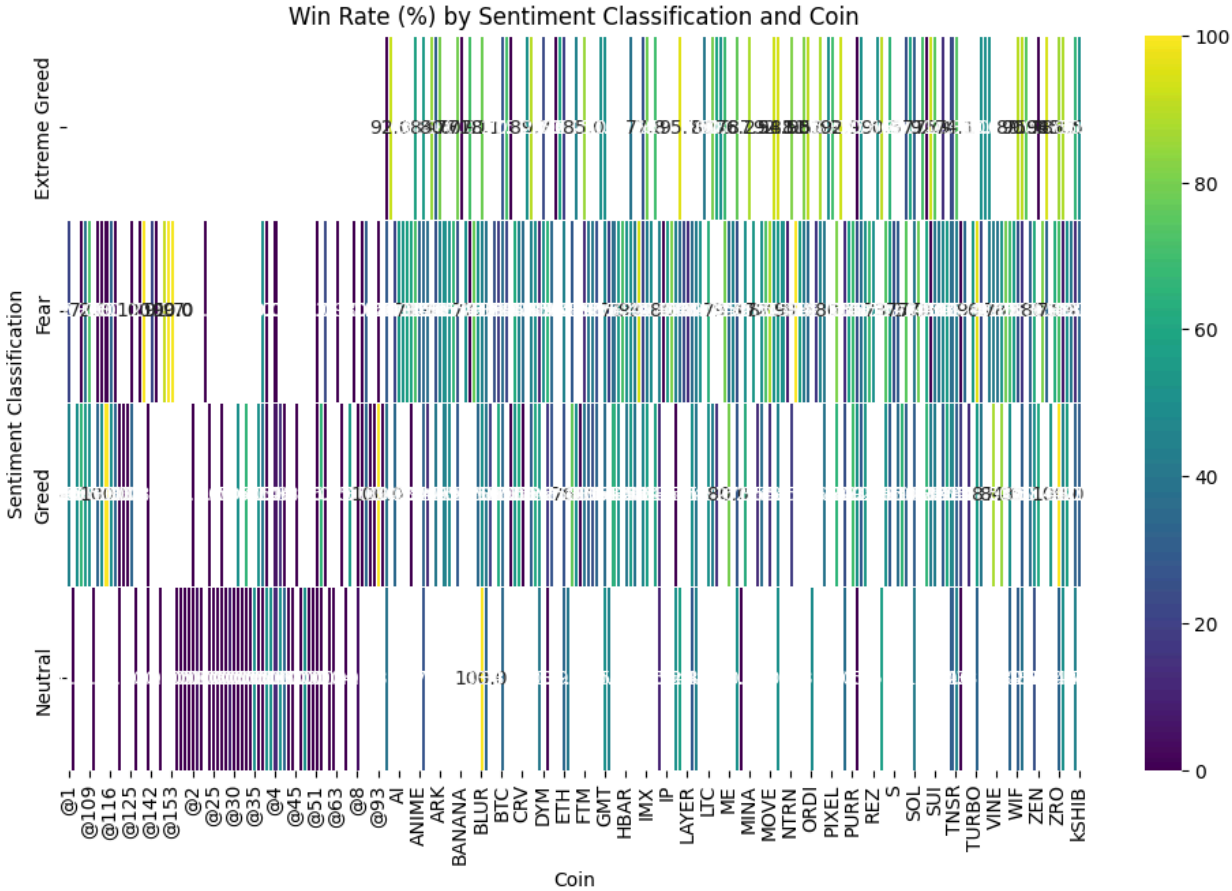
- **Trade Volume Heatmap:** Highlighting that trading frequency increases during sentiment shifts, which often correlates with local market tops or bottoms.



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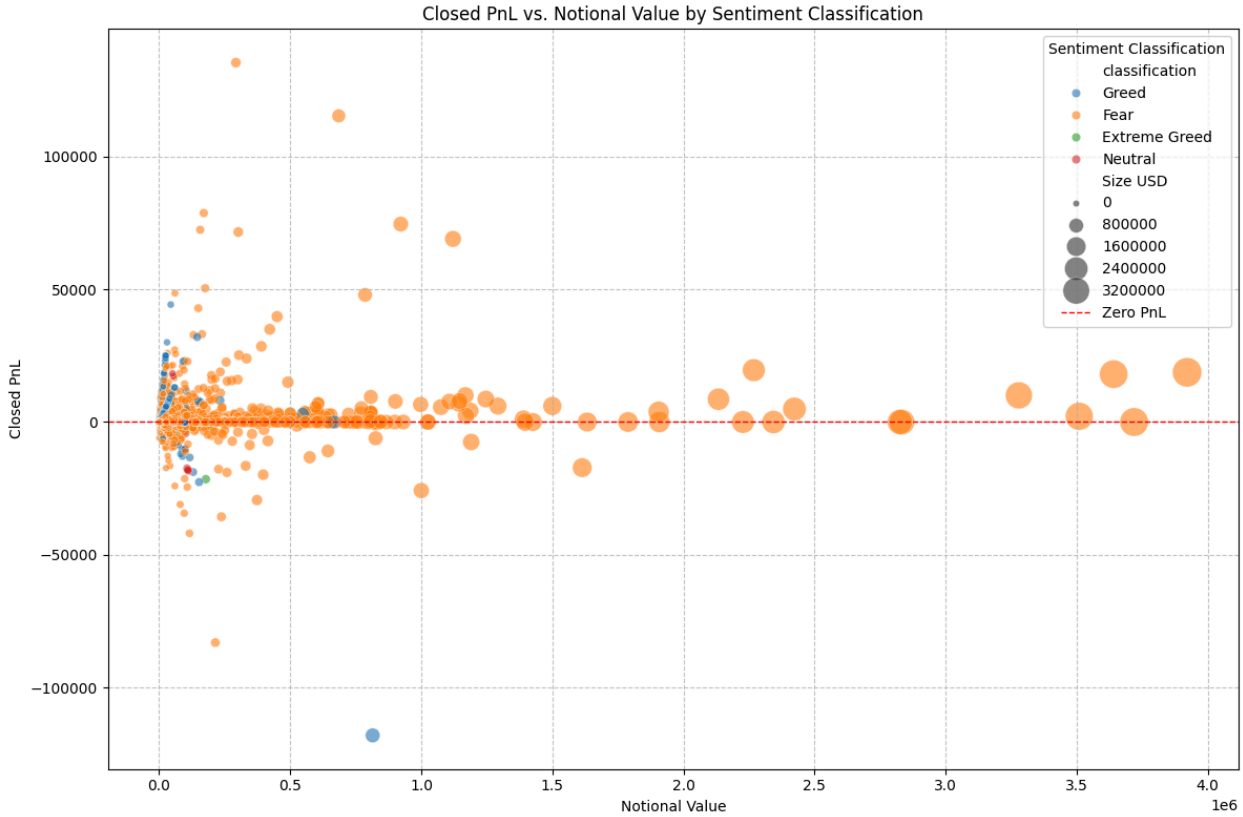
- **Scatter Plot (PnL vs. Notional):** Confirmed that higher capital deployment (Notional Value) during sentiment extremes led to the largest PnL variances, both positive and negative.



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6. Actionable Trading Insights

1. **Avoid the Middle:** Avoid heavy capital deployment during "Neutral" sentiment, as it has the lowest statistical edge for both win rate and profit factor.
2. **Focus on BTC for Large Moves:** Utilise "Greed" periods to look for outlier BTC trades, as winners in this environment are significantly larger than losses.
3. **Emulate Smart Money Hedging:** During "Extreme Greed," successful traders do not just go "Long"; they balance their BUY/SELL ratios to protect against the high probability of a trend reversal.